Unnath Chittimalla

RTL Design · FPGA Implementation · Computer Architecture · Hardware-Aware ML

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Education

International Institute of Information Technology Bangalore (IIIT-B) **Jul 2023 – Jul 2028**

Integrated M.Tech (Dual Degree B.Tech + M.Tech) in Electronics and Communication Engineering *CGPA: 3.49 / 4.00*

- **Relevant Coursework:** VLSI Design, Computer Architecture & Processor Design, Digital Design, Signals & Systems, Embedded Systems & Robotics, Control Systems & LQR, Data Structures & Algorithms.

Technical Skills

HDLs & Computer Architecture: SystemVerilog, Verilog, VHDL, Vitis HLS, Custom GPU ISA, Pipelining
EDA & FPGA Tools: Xilinx Vivado (OOC Synthesis, P&R, STA, VIO/ILA), Cadence Virtuoso (180nm GPDK), SkyWater 130nm PDK
Simulation, Power & Verification: Verilator, Icarus Verilog, cocotb, gem5, DRAMsim3, DRAMPower, LTspice, NGspice, C++ Golden Models
Scripting, Software & DSP/ML: Python, Tcl (EDA Batch Scripting), C++, C, Rust, Kotlin, Swift, MATLAB, PyTorch, TensorFlow, Fixed-Point Quantization, Git, Linux

Research Experience & IEEE Publications

Undergraduate Researcher, IIIT Bangalore **Jun 2025 – Present**

Advisor: Dr. Madhav Rao (Professor, ECE Department, IIIT Bangalore) *Bangalore, India*

- **First Author (IEEE EMBS BHI 2026):** *"Adaptive Lead-Field Residual Learning for Sparse-to-Dense EEG Reconstruction"* (*GitHub*). Formulated EEG spatial super-resolution using a Physics-Informed Residual Network with learnable Boundary Element Method (BEM) conductivity calibration (G_{eff}) and forward-consistency regularization ($\mathcal{L}_{\text{fwd}}$).
- Benchmarked 9 reconstruction paradigms across 5-, 10-, and 15-channel layouts under Leave-One-Subject-Out (LOSO) CV on BCI Competition IV-2a; reduced reconstruction error from 9.69 μV (static BEM) to **2.215 μV RMSE** ($R^2 = 0.955$, $r = 0.982$), enabling **77% wireless sensor bandwidth/power reduction** (5 → 22 channels) without structural MRI.
- **Co-Author (IEEE EMBS BHI 2025, Atlanta):** *"EEG-Based Surface EMG Reconstruction Using Deep Sequence Learning for Upper Limb Motor Activity"* (DOI: 10.1109/BHI67747.2025.11269463; *Code*). Learned direct cortico-muscular mappings from non-invasive EEG via signal preprocessing plus spatio-temporal modeling, replacing dual-modality EEG-EMG systems with a single EEG modality. Implemented `RawSeq2SeqNet` (CNN-LSTM, Butterworth/Hilbert DSP, CSP, MSE + SoftDTW loss) with **2.11 ms real-time inference latency** ($r = 0.795$).

Hardware Design, Computer Architecture & EDA Projects

DDR4 PDN Decoupling & EDA Automation Flow **Apr 2026 – Jun 2026**

gem5, DRAMsim3, LTspice, Python *GitHub*

- Built a **7-stage EDA automation pipeline** coupling **gem5**, **DRAMsim3**, and **DRAMPower** to convert DDR4-3200 command traces (Rowhammer, streaming, random) into SPICE `.pwl` waveforms for simultaneous switching noise (SSN) on VDDQ/VTT rails.
- Developed a batch-LTspice **Genetic Algorithm (GA)** optimizing an 86-capacitor network (51 VDDQ, 35 VTT) for an `LTM4632` buck regulator, suppressing anti-resonance spikes up to **1.6 GHz** within JEDEC ripple limits (±60 mV VDDQ / ±40 mV VTT) under **0.948 nH** via inductance.

OBC Distributed-Arithmetic DSP Core & HLS MLP **Oct 2025 – Jun 2026**

Verilog, VHDL, Vitis HLS, Artix-7 *GitHub*

- Designed a multiplier-less Q15 fixed-point inner-product DSP core using **Distributed Arithmetic (DA)** with **Offset Binary Coding (OBC)** to cut **LUT usage by 50%**, plus a 4-stage pipelined **Baugh-Wooley multiplier** reducing critical-path logic delay by **2.1**× (1.473 ns → 0.704 ns, 5 → 2 logic levels, 166.7 MHz).
- Implemented a 16-bit fixed-point quantized neural network (784 × 64 × 10 MLP with hardware `tanh`/ReLU) in **Vitis HLS & VHDL**, validated at-speed on board on an **Artix-7 (Basys 3)** FPGA using **Vivado VIO and ILA** on-chip logic analyzers.

Hybrid CMOS-ReRAM Controller & Memristor Solver **Nov 2025 – Jan 2026**

SkyWater 130nm, Virtuoso *GitHub*

- Designed (RTL and simulation) a **SkyWater 130nm**-targeted Hybrid CMOS-ReRAM memory controller with an AXI-Lite-style Ready/Valid interface, RTL Program-and-Verify FSM, 1.8V-to-3.3V level shifters, and 1T1R bitline pulse-shaping.
- Simulated an $\mathcal{O}(1)$ analog **Memristor Crossbar Matrix Equation Solver** in **Cadence Virtuoso (GPDK 180nm)**, analyzing conductance programming and peripheral overhead (99%+ numerical accuracy in simulation).

TinyGPU: Minimal GPU in SystemVerilog (based on a public learning project) **Dec 2025 – Aug 2026**
SystemVerilog, Icarus Verilog, cocotb *GitHub | Video walkthrough*

- Studied and built a minimal multi-core GPU in **SystemVerilog**, following the public adam-maj/tiny-gpu learning project: dispatcher, per-core ALU/LSU pipelines, memory controller and a custom ISA; simulated matrix addition and multiplication kernels with **cocotb** (RTL design and simulation only).

IEEE 802.11ad Layered Min-Sum LDPC Decoder (RTL & Simulation) **Feb 2026 – Apr 2026**
SystemVerilog, Verilator/Icarus, C++ golden model *GitHub*

- SystemVerilog RTL and functional simulation of a layered min-sum decoder; work in progress.

spiceRL: Buck Converter Sizing RL Environment **2026**
Python, OpenEnv, LTspice/NGspice *GitHub*

- Small OpenEnv RL environment for sizing buck converter MOSFETs.

Embedded Systems, Signal Processing & Product Engineering

IMU-Only 2D Dead Reckoning with Sensor Fusion **Sep 2025**
Python, DSP, Kalman Filtering *GitHub*

- Mitigated accelerometer/gyroscope integration drift for 2D dead reckoning by designing FFT-informed **Butterworth filters**, **Kalman filters**, and a feature-driven Random Forest velocity regressor, achieving **<2.5% distance error** against GPS ground truth.

GoalGuard Platform (Co-Founder) **Jul 2025 – Present**
Kotlin, Android, System Services *Website · Play Store*

- Co-founded a privacy-first productivity app on Google Play with **20,000+ downloads across 20+ countries**; selected to attend **Y Combinator Startup School India** (top 2,000 from 20,000+ applicants).

relayBrain: Coordination Layer for AI Coding Agents (Team Project) **Jun 2026 – Oct 2026**
JavaScript, Node.js, FalkorDB *GitHub · Website*

- Contributor (17 commits) to a team project led by @prakrititz: shared file locks, live patch sync, and a FalkorDB memory graph so multiple AI coding agents (Cursor, Claude Code, Copilot, Codex) can edit one repository without overwriting each other.

Circadian Lighting Firmware (e-Yantra, IIT Bombay) **Apr 2024 – Feb 2025**
ESP32, Embedded C, Firmware *GitHub*

- Developed embedded C firmware and mathematical circadian spectrum models on ESP32 microcontrollers to synchronize multi-node smart lighting arrays across heterogeneous hardware protocols.

Honors, Hackathons & Academic Leadership

- **Stage 2 Winner, NCIIPC & Startup India AI Grand Challenge (2026):** Team Coders@IIITB, PS-11 Hyperspectral Anomaly Detection; implemented Robust PCA and GAN-BWGNN pipelines (*Code*).
- **Top 10 Finalist, Meta Pragati AI for Impact (2025, New Delhi):** Among **10,000+ teams**; worked on a low-latency multilingual RAG system.
- **Top 11 Finalist, ELCIA Tech Summit Sensor Challenge (2025):** Team 515 Labs; implemented a **quantized Random Forest model** for on-device motion artifact filtering (*Code*).
- **1st Place (Among 5,000+ Participants), ICDEC VDVWC Challenge (2024):** Ranked 1st with a modified YOLOv10 for adverse-weather vehicle detection (*Code*).
- **3rd Place, Aptos SuperMove Tour (2024, Bangalore):** Built decentralized smart contracts in the Move language for IPFS media provenance (*Code*).
- **Teaching Assistant (Signals & Systems) & Core Member (Zense Tech Club), IIIT Bangalore:** Conducted tutorials, labs, and evaluations for 150+ ECE undergrads; helped run technical workshops.